DIGITAL INNOVATIONS · LEARNING & DEVELOPMENT

From Classroom to Crisis Room

How AI-powered simulation gaming solves the enterprise upskilling challenge — at scale, in context, and with measurable outcomes.

Presented by	Date	Classification	Status
Digital Innovations L&D	February 2026	Internal — Executive Review	Proof of Concept Complete

OVERVIEW

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What We'll Cover Today

- **The Problem:** Why the AI skills gap is different from every upskilling challenge before it
- **The Failure Mode:** Why traditional classroom training can't solve it
- **The Solution:** Experiential AI simulation gaming — learn by doing, not listening
- **Dual Learning Design:** AI fluency + real business problem-solving in a single experience
- **Feature Walkthrough:** What the platform does and how it works

> *"The best way to understand what AI can do is to use it — not to sit in a room and be told about it."*
>
> **— The core design principle behind this solution**

🎯 **The Bottom Line**

We built a working prototype of an AI-powered, multiplayer business simulation game that teaches AI fluency through authentic, high-stakes problem-solving. It is scaleable, customizable, and ready for pilot.

- **POC vs. Full Build:** What we proved, what we're still building toward
- **The Journey:** How we built a fully functioning prototype in days using AI-assisted development
- **Business Case & Next Steps**

THE PROBLEM

3 The AI Skills Gap Is Not a Training Problem — It's a Transformation Crisis

85%
of jobs will be transformed by AI within 5 years, yet most employees have never used an AI tool in a meaningful work context

3×
faster pace of AI adoption vs. any prior digital transformation — companies have 18 months before the gap becomes a competitive liability

$4.6T
annual productivity value on the table — enterprises that deploy AI-fluent workforces outperform peers by 2.5× on innovation metrics

WHAT EXECUTIVES NEED THEIR TEAMS TO DO

- **Use AI as a co-analyst** — query data, synthesize documents, build rapid hypotheses
- **Apply AI to real work** — not demos, not sandboxes, not theory
- **Think in "AI-first" workflows** — when to use it, how to validate outputs, where it fails

⚠️ **The Paradox**

You cannot teach someone to use AI by talking to them about AI.

Conceptual training creates awareness, not capability. Capability comes from using the tool under realistic pressure — with real consequences for getting it wrong.

No existing enterprise training platform delivers this.

- **Build confidence fast** — AI anxiety is a real productivity blocker at every level

THE PROBLEM

4 Why Traditional In-Classroom Training Can't Solve This

Dimension	Traditional Classroom / eLearning	Why It Fails for AI Upskilling
📏 **Scale**	20–30 learners per session max. Instructor dependency. Scheduling bottlenecks across time zones and shifts.	A global workforce cannot be trained one classroom at a time. By the time cohort 1 finishes, the AI landscape has changed.
💤 **Engagement**	Passive learning. Retention drops 70% within 24 hours (Ebbinghaus Forgetting Curve). Completion rates for eLearning modules average 15%.	Watching a demo of Claude doesn't make someone fluent in Claude. Passive observation of AI creates false confidence without real skill.
🌍 **Context**	Generic scenarios. Learners must mentally translate generic content to their actual job context — a leap most never make.	AI skills are deeply context-dependent. Using AI for supply chain analysis is different from using it for HR. Generic training wastes time.

⏱️ **Time to Impact**	Courses take 4–8 hours. Change programs take months to design. ROI hard to measure. Behavior change lags by 3–6 months.	The competitive window is now. A solution that takes 6 months to design and deploy is already obsolete.
🔁 **Reusability**	Instructor-led content ages rapidly. Must be rebuilt as AI tools evolve. Single-use curriculum investment.	AI tools update monthly. Training content must be modular and swappable, not monolithic.

The fundamental gap: Traditional training teaches *about* AI. We need training that teaches *through* AI.

The solution must embed AI tool usage as the mechanism of learning — not as the subject matter on a slide.

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Our Solution

AI-Powered Simulation Gaming — learn by solving a real crisis, guided by an AI co-pilot, alongside a teammate, against the clock.

THE SOLUTION

6 Introducing: AI-Powered Business Simulation Gaming

Participants are placed inside a live, time-pressured business crisis — a supply chain failure, a cybersecurity incident, a product recall, a regulatory breach — and must solve it using AI tools. The simulation runs in real time, adapts to player behavior, and delivers a personalized debrief that closes the learning loop.

THE PLAYER JOURNEY

- **Lobby:** Two players assigned to complementary roles — each holds different information and must collaborate to solve the puzzle
- **Investigation Phase:** Players use AI tools to analyze documents, surface patterns, and discover clues hidden across a rich file library
- **War Room:** A shared collaboration space where teams pool findings in real time — modeled on how AI-enabled teams actually work

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Concurrent learning objectives in a single session

LEARNING OBJECTIVE 1

🤖 AI Tool Fluency

Participants must use AI (Claude, ChatGPT, Copilot) to analyze documents, synthesize data, and form hypotheses — or they won't solve the problem. AI usage is the mechanism, not the subject.

LEARNING OBJECTIVE 2

🏭 Real Business Acumen

The scenario is a realistic model of an actual business problem — supply chain cascading

- **AI Coach:** A live AI advisor monitors progress and injects contextual hints without giving away the answer — adaptive, personalized guidance
- **Escape Submission:** Teams converge on a root cause and recommended action — the moment of hypothesis formation under pressure
- **AI Evaluation + Debrief:** The AI grades the solution, explains the correct answer, and generates a customized debrief with real-world implications

failure, cybersecurity incident response, regulatory crisis. Teams learn domain knowledge while learning AI.

PLATFORM FEATURES

7 Platform Feature Architecture — What It Does and How

Asymmetric Role Design

Each player holds unique information — forcing communication, eliminating "one genius solves it" dynamics. Mirrors real cross-functional AI teamwork.

Rich Document Investigation

10–20 realistic business documents per scenario — spreadsheets, emails, audit reports, supplier profiles. Formatted for AI-assisted analysis.

Gate Questions

Key documents are locked behind comprehension questions — ensuring players engage with content rather than speed-clicking. Includes hints on wrong answers.

Live War Room

Real-time shared collaboration channel. Players post findings, AI coach interjects. Mirrors Slack/Teams-based AI-augmented team workflows.

Adaptive AI Coach

Monitors player behavior in real time. Fires contextual hints when players are stuck, at checkpoints, or after specific events. Never gives away the answer.

Countdown Timer + Pressure

Configurable time limit creates urgency that mirrors real crisis decision-making. Pressure reveals how people actually use (or don't use) AI under stress.

AI Evaluation Engine

Submissions scored against a rubric using an AI evaluator. Partial credit for correct reasoning even with imperfect answers. Calibrated, consistent grading.

Personalized Debrief

AI generates a full post-game debrief: what happened, what the team got right, what was missed, real-world business implications, and AI tool usage lessons.

Moderator Dashboard

Facilitators observe the session in real time — see both players' file activity, clue discoveries, War Room messages, and session status. Full reset capability.

SCALE & DESIGN

8 Why This Is a Scaleable Solution — The Design Decisions That Matter

ARCHITECTURE FOR SCALE

- **No instructor dependency:** The AI coach replaces the human facilitator for hint delivery, evaluation, and debrief. One facilitator can oversee dozens of simultaneous sessions.
- **Simultaneous sessions:** Any number of pairs can run concurrently. A 500-person org can be trained in a single day with minimal coordination overhead.
- **Async-ready:** Platform is designed for remote, distributed teams. No room

SCALABILITY VS. TRADITIONAL TRAINING

1 day

to train 500 employees in concurrent sessions vs. 25 weeks of classroom cohorts for the same population

~$18

estimated cost per learner at scale (AI API + infrastructure) vs. $200–$800 per learner for facilitated in-person training

90 min

total learning experience per session — including debrief — vs. full-day classroom requirement for equivalent content depth

	booking, no travel, no time zone synchronization required for participants.
Scenario library:	New scenarios can be authored and deployed without rebuilding the engine. One engine, infinite scenarios — supply chain, cybersecurity, HR crisis, finance, operations.
LMS integration path:	Architecture supports SCORM/xAPI export and SSO integration for embedding into Workday Learning, Cornerstone, LinkedIn Learning, or internal portals.

PROOF OF CONCEPT

9 The POC: "Apex Meltdown" — What We Built and What It Proves

SCENARIO: THE CASCADE

A supply chain crisis simulation in which two participants — a **Procurement Analyst** and an **Operations Manager** — must uncover a cascading component shortage threatening a 48-hour production line stoppage. Each player holds different documents. Neither can solve it alone.

WHAT THE POC DEMONSTRATES

- **Full game loop works end-to-end:** lobby → join → investigate → collaborate → submit → evaluate → debrief
- **Real-time AI coaching functions:** adaptive hints fire based on player behavior; coach monitors progress and intervenes contextually
- **AI evaluation is credible:** Submissions scored with nuance — partial credit, missed elements, correct reasoning flagged
- **Dual-persona design creates** Players must communicate or fail —

TECH STACK

Next.js 14
FastAPI
WebSocket
Redis
Anthropic Claude API
Python 3.11
TypeScript
Tailwind CSS

Real-time multiplayer backend · AI-native architecture · Production-grade component library

🎯 What This POC Proves to Stakeholders

The concept is not theoretical. A fully functional, playable experience exists today. The scenario, the AI coaching, the collaborative

authentic collaboration:	information asymmetry works as designed
Moderator observability is functional:	Full session visibility, reset, and override capability in place
Enterprise branding is production-ready:	Toyota/DI design system applied; easily reskinned per business unit

mechanics, and the debrief — all working, all demonstrable, all ready for pilot evaluation.

POC VS. FULL BUILD

10 POC Boundaries vs. Enterprise-Grade Full Build

Capability	POC (Current State)	Full Build (Target State)
Scenarios	1 hard-coded scenario (supply chain). New scenarios require developer work on JSON config files.	No-code scenario authoring tool. L&D teams build and publish new scenarios in 2–4 hours. Scenario library grows continuously.
Player Count	2-player format only. Persona count is fixed in the scenario definition.	2–8 players per session. Variable team sizes. Solo mode for self-paced learning. Configurable per scenario.
Authentication	URL-based session sharing. No user accounts, no identity management.	SSO / SAML integration. Ties to existing enterprise IdP (Okta, Azure AD). Pre-assigns roles based on org data.
Analytics	Session data ephemeral (Redis, 24-hour TTL). No historical tracking, no	Full learning analytics dashboard. Cohort comparisons, completion rates, score trends, AI

	aggregate reporting.	tool usage patterns, manager-level rollup views.
LMS Integration	Standalone. No connection to existing learning systems.	SCORM/xAPI compliant. Embeds into Workday Learning, Cornerstone, LinkedIn Learning, custom portals.
AI Model Choice	Anthropic Claude only. API key managed manually.	Model-agnostic. Supports Claude, GPT-4o, Gemini. AI provider swap without scenario rebuild.
Facilitation Toolkit	Moderator panel functional but no facilitator playbook, run guide, or pre/post-session materials.	Complete facilitator toolkit: run guide, participant briefing, debrief facilitation framework, customization guide.
Infrastructure	Local/dev deployment. No cloud hosting, CDN, load balancing, or HA setup.	Cloud-native deployment (AWS/Azure). Auto-scaling, global CDN, 99.9% uptime SLA, GDPR-compliant data handling.

THE JOURNEY

11 How We Got Here: Claude Code as Co-Developer

PHASE 1 · CONCEPT

Idea → Architecture in Hours

Described the learning experience to Claude Code. It generated a full-stack architecture proposal (Next.js + FastAPI + WebSocket + Redis), explained the trade-offs, and scaffolded the initial project structure.

PHASE 2 · CORE BUILD

Working Game Loop in Days

Multiplayer WebSocket engine, real-time state sync, Redis session management, file investigation system, gate questions, War Room — all built through iterative Claude Code sessions with natural language instruction.

PHASE 3 · AI INTEGRATION

AI Coach + Evaluator

Adaptive coach agent that monitors player state and fires contextual hints. AI evaluator that grades submissions with nuance and generates personalized markdown debriefs. All wired to Anthropic API.

PHASE 4 · DESIGN

> *"Traditional development of this scope — full-stack multiplayer game with AI integration — would take a team of 3–4 developers 3–4 months. This was built in days."*

— Demonstrated in this project

The Meta-Point

The platform teaches participants to use AI for complex problem-solving. The platform *itself* was built using AI for complex problem-

Enterprise Reskin

Toyota-inspired enterprise design system applied. DI brand assets integrated. Countdown timer, markdown rendering, moderator panel, and "New Session" flow added through feedback-driven iteration.

PHASE 5 · HARDENING

Bug Fixes & Polish

Python version compatibility issues resolved, Redis startup automated, npm dependency management fixed, WebSocket reconnection handled. All through conversational debugging with Claude Code.

solving. This project is its own proof of concept for AI-fluent work.

⚡ Development Approach

Claude Code acted as senior architect, full-stack developer, debugger, and code reviewer — all within a single conversational session. Human direction + AI execution = multiplicative output.

STRATEGIC INSIGHT

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The Meta-Learning Opportunity: This Project IS the Lesson

Learners use AI to solve a realistic crisis scenario — experiencing AI's analytical power firsthand in a consequence-safe environment

AI coaches the learners in real time — adapting to their behavior, providing guidance, evaluating their work, and closing the learning loop

AI built the platform — demonstrating that AI-fluent teams ship faster, cheaper, and at higher quality than traditional development models

> *"Every touchpoint in this solution — the creation, the delivery, and the learning — is powered by AI. We are not pitching a product that teaches about AI. We are demonstrating what AI-native work looks like."*

For Participants

They leave having used AI to solve a real problem. Not having watched a

For the Organization

A platform built with AI to teach AI — the procurement cost and

demo. Not having read a slide. They leave with a memory of the moment AI helped them see something they couldn't see alone — and that memory transfers to their actual job.

development timeline are themselves evidence of what AI-fluent teams can accomplish. This pitch deck is a case study before the product even launches.

BUSINESS CASE

13 The Business Case: Why the Investment Makes Sense Now

COST COMPARISON (PER 100 LEARNERS)

Approach	Estimated Cost	Time Required
In-Person Workshop (facilitated)	$40,000–$80,000	5 days of scheduling + 1 day delivery
eLearning Module (custom)	$30,000–$60,000 dev + $5K/yr maintenance	3–6 months development
External Training Vendor	$50,000–$150,000+	Months of procurement + scheduling
AI Simulation Platform (full build)	~$1,800 AI API costs + platform investment	1 day to train 100 concurrent learners

STRATEGIC VALUE DRIVERS

- **Reusable engine:** One platform investment supports unlimited scenario variations across all business units and functions
- **Speed to capability:** 90-minute immersive experience vs. multi-day programs — faster behavior change, faster ROI

PIL

- **Measurable outcomes:** Every session generates quantified data — score, clues found, time to solution, AI usage patterns

NEXT STEPS

14 What We're Asking For — and What Happens Next

IMMEDIATE ACTIONS (30 DAYS)

- **Executive sponsor approval** to proceed to pilot phase — formal sign-off on resource allocation and target population
- **L&D team alignment** — identify pilot cohort, assign facilitators, align with existing training calendar
- **IT/Security review** — brief cloud deployment, SSO integration requirements, data residency standards
- **Scenario co-design session** — work with business unit leaders to identify the 2–3 highest-priority scenario themes beyond supply

PHASED ROADMAP

Phase 1 · Now

POC Live Pilot — run 30-person pilot with current POC. Collect data. Validate hypothesis. 2–4 weeks.

Phase 2 · Q2

Full Build v1.0 — scenario authoring, SSO, analytics, cloud deployment. 60–90 days. Ready for enterprise rollout.

Phase 3 · Q3

Scenario Library Expansion — 5+ scenarios across functions. Cybersecurity, HR, Finance, Operations. Co-authored with BU leaders.

Phase 4 · Q4

Platform as Service — offer to enterprise clients as a branded, white-label upskilling solution. New revenue line for DI.

chain (e.g., cybersecurity incident, product recall, regulatory audit)

FULL BUILD INVESTMENT

- **Scenario authoring tool** — no-code builder for L&D teams to create and publish new scenarios independently
- **SSO + LMS integration** — plug into existing enterprise learning infrastructure
- **Analytics dashboard** — cohort-level reporting for L&D leadership and business unit sponsors
- **Cloud deployment + SLA** — production infrastructure with uptime guarantee and auto-scaling
- **Facilitator toolkit** — run guide, participant materials, debrief framework

DIGITAL INNOVATIONS · LEARNING & DEVELOPMENT

The Question Isn't Whether to Upskill on AI.

The question is whether you want your team to experience the power of AI firsthand, in a consequence-safe environment designed to create lasting behavior change — or to sit through another slide deck about it.

✓ Built

Working POC. Playable today. Full game loop end-to-end.

✓ Proven

Concept validated. AI coaching and evaluation work. Enterprise design ready.

→ Ready

Pilot-ready now. Full build in 60–90 days. Scale in 6 months.

Contact	Platform	Built with	Status
Digital Innovations L&D Team	Apex Meltdown / The Cascade	Claude Code (AI-Assisted Development)	POC Complete · Pilot Ready